Software Requirements Specification: Personalized Learning Path Recommender for Online Education

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Version 1.0 draft, structured after IEEE 830 / ISO/IEC/IEEE 29148. It specifies what the system must do; the companion PRD explains why.

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Personalized Learning Path Recommender for Online Education. It is written for developers, testers, instructors and project evaluators, and is the baseline for design, implementation and testing.

1.2 Scope

The system analyzes a learner's quiz performance and engagement patterns and recommends a personalized sequence of learning content.

- **In scope:** knowledge model per learner and topic; recommendation engine; adaptive quizzing; learner and instructor progress dashboards; A/B testable recommendation strategies; basic account, role and content-tagging management.
- **Out of scope (v1):** authoring course content, live video or tutoring, payments and certificates, native mobile apps, free-text answer grading.

1.3 Definitions and abbreviations

Term	Meaning
Mastery score	Estimate between 0 and 1 of how well a learner knows a topic
Knowledge model	Component that stores and updates mastery per learner and topic
Learning path	Ordered sequence of content items recommended to a learner

Term	Meaning
Strategy	A recommendation algorithm behind a common interface (rules, embeddings, collaborative filtering)
Arm	One variant of an A/B experiment, tied to one strategy and its parameters
Cold start	A learner or content item with no history yet
Adaptive quiz	Quiz whose next question difficulty depends on previous answers
SRS / PRD	Software requirements specification / product requirements document
RBAC	Role-based access control

1.4 References

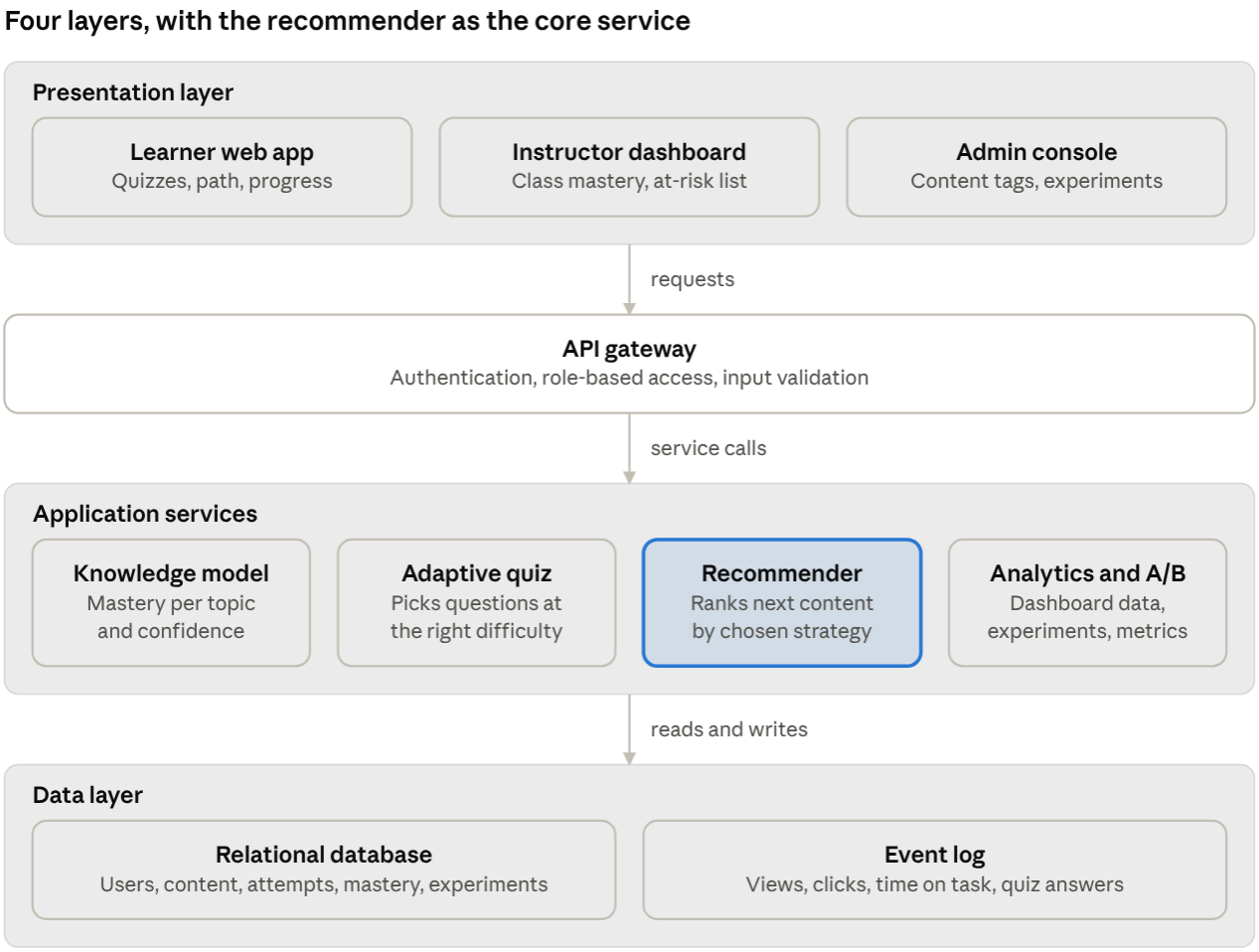
- IEEE 830-1998 and ISO/IEC/IEEE 29148:2018, requirements specification standards.
- Companion document: PRD for the Personalized Learning Path Recommender (goals, personas, user stories US-01 to US-12).

1.5 Document conventions

Requirements use "shall" and carry unique IDs: FR for functional, NFR for non-functional. Priority is M (must), S (should) or C (could). Default parameter values are initial settings that administrators can change.

2. Overall description

2.1 Product perspective



system architecture · 4 layers

The system is a self-contained web application. Clients talk only to the API gateway; services never expose themselves directly. The recommender sits behind a strategy interface so that algorithms can be swapped and A/B tested without changing clients.

2.2 Product functions

- Track per-topic mastery for each learner.
- Recommend a personalized sequence of content.
- Adapt quiz difficulty from performance.
- Show progress to learners and instructors.
- Run A/B tests across recommendation strategies.

2.3 User classes

Class	Description	Technical skill
Learner	Takes quizzes and follows the recommended path	Basic web use
Instructor	Monitors class progress and at-risk learners	Basic web use
Administrator	Tags content, configures strategies and experiments	Moderate

2.4 Operating environment

Modern desktop and mobile browsers (current and previous major versions of Chrome, Firefox, Safari, Edge); server side runs on a Linux host or container platform.

2.5 Design and implementation constraints

- Every recommendation must be explainable with a one-line reason.
- Learner data is personal data; access follows role-based rules.
- Recommendation strategies must implement one common interface.

2.6 Assumptions and dependencies

- A course catalog of lessons and quizzes exists or is seeded.
- Content can be tagged by topic, difficulty and prerequisites.
- Learners authenticate with an account before any tracking happens.

3. External interface requirements

ID	Interface	Requirement
EI-1	User interface	A responsive web interface shall provide: login, course home with the recommended path, quiz player, learner dashboard, instructor dashboard and admin console.
EI-2	User interface	The interface shall meet WCAG 2.1 level AA and work from a 360 px wide screen up.
EI-3	Software: API	Clients shall use a versioned REST API exchanging JSON; every endpoint except login and registration shall require an authenticated token.

ID	Interface	Requirement
EI-4	Software: database	The services shall access a relational database for structured data and an append-only event log for interaction events.
EI-5	Software: LMS (optional)	If the system is embedded in an existing learning platform, it shall accept launches through the LTI standard. Decision pending in the PRD open questions.
EI-6	Hardware	No special hardware. Clients need a modern browser; the server runs on a standard Linux host or container.
EI-7	Communication	All traffic shall use HTTPS with TLS 1.2 or later.

4. Functional requirements

Each module lists its requirements, priority and the PRD user story it traces to.

4.1 Accounts and access (FR-1xx)

ID	Requirement	Pri	Traces to
FR-101	The system shall let a user register and log in with email and password.	M	All
FR-102	The system shall support three roles: learner, instructor, administrator.	M	US-07, US-09
FR-103	The system shall restrict each role to its own functions; a learner sees only their own data and an instructor only their own courses.	M	US-07, US-08
FR-104	The system shall record the learner's consent before tracking activity.	M	Privacy

4.2 Content management (FR-2xx)

ID	Requirement	Pri	Traces to
FR-201	An administrator shall create, edit and retire courses, topics, content items and quiz questions.	M	US-09

ID	Requirement	Pri	Traces to
FR-202	Each content item shall carry topic, difficulty (1 to 5), type, estimated duration and prerequisite topics.	M	US-09
FR-203	The system shall report content items missing required tags.	S	US-09

4.3 Knowledge model (FR-3xx)

ID	Requirement	Pri	Traces to
FR-301	The system shall keep a mastery score (0 to 1) and a confidence value for every learner-topic pair.	M	US-02
FR-302	The system shall update mastery after each graded attempt, before the next recommendation or question is served.	M	US-02, US-04
FR-303	The system shall initialize mastery from the diagnostic quiz, or from a default prior when none exists.	M	US-01
FR-304	The system shall weight recent attempts more than older ones.	S	US-02
FR-305	The system shall decay mastery for topics not practiced for a configurable period.	S	US-03
FR-306	The system shall label each topic weak, developing or strong using configurable thresholds (default: weak below 0.40, strong 0.80 or above).	M	US-02
FR-307	The system shall log engagement events (content opened, completed, skipped, time on task, quiz answered) and derive per-learner completion rate, skip rate and average session time.	M	US-03, US-06

4.4 Recommendation engine (FR-5xx)

ID	Requirement	Pri	Traces to
FR-501	On request, the system shall return an ordered list of the next N content items for a learner (default N = 5).	M	US-03
FR-502	The system shall not recommend items already completed, unless a review is due for a weak topic.	M	US-03
FR-503	The system shall not recommend an item whose prerequisite topics are still weak.	M	US-03
FR-504	The system shall rank candidates by the weighted score defined in section 7.2.	M	US-03
FR-505	The system shall attach a one-line, human-readable reason to every recommendation.	M	US-03
FR-506	The system shall implement every strategy behind one common interface: rule-based (MVP), embeddings plus rules, collaborative filtering.	M	US-10
FR-507	The embeddings strategy shall compute content similarity from title, description and tags.	S	US-03
FR-508	The collaborative strategy shall use paths that similar learners completed successfully.	C	US-03
FR-509	The system shall let a learner dismiss a recommendation and shall not re-suggest it for a configurable period.	S	US-05
FR-510	For a learner with no history, the system shall fall back to diagnostic results, or to entry-level content.	M	US-01

4.5 Adaptive quizzing (FR-6xx)

ID	Requirement	Pri	Traces to
FR-601	The system shall offer a diagnostic quiz covering every topic when a learner joins a course.	M	US-01

ID	Requirement	Pri	Traces to
FR-602	The system shall choose the next question's difficulty from current mastery and recent answers, stepping up after consecutive correct answers and down after consecutive incorrect ones (section 7.3).	M	US-04
FR-603	The system shall not repeat a question within one quiz session.	M	US-04
FR-604	The system shall store every attempt with correctness, time taken and timestamp.	M	US-02
FR-605	The system shall end a quiz when mastery confidence passes a threshold or a maximum length is reached.	S	US-04
FR-606	The system shall recalibrate question difficulty from aggregate response data.	C	US-04

4.6 Dashboards (FR-7xx)

ID	Requirement	Pri	Traces to
FR-701	The learner dashboard shall show mastery per topic, recommended next items, completion and time spent.	M	US-02, US-06
FR-702	The instructor dashboard shall show a class mastery heatmap by topic, average completion and an at-risk learner list with configurable criteria.	M	US-07
FR-703	An instructor shall open one learner's attempt and activity history.	S	US-08
FR-704	Dashboards shall filter by course, topic and date range.	S	US-07
FR-705	A user shall export their visible report as CSV or PDF.	C	US-12

4.7 A/B testing (FR-8xx)

ID	Requirement	Pri	Traces to
FR-801	An administrator shall create an experiment with a name, arms (strategy and parameters), allocation split, primary metric and start and stop dates.	M	US-10
FR-802	The system shall assign each learner to one arm at random and keep that assignment fixed for the experiment.	M	US-10
FR-803	The system shall log each recommendation with its experiment and arm.	M	US-10
FR-804	The system shall compute per arm: recommendation acceptance rate, 8-week completion rate and post-test score gain.	M	US-10
FR-805	The system shall report significance and a confidence interval for each metric.	S	US-10
FR-806	The system shall pause an experiment automatically if a guardrail metric worsens beyond its limit.	S	US-10
FR-807	An administrator shall switch a strategy on or off without redeploying.	S	US-11

5. Data requirements

5.1 Entities

Entity	Key attributes	Belongs to / links
User	user_id (PK), email, role, consent_at	Parent of Attempt, MasteryScore, InteractionEvent, Recommendation, ExperimentAssignment
Course	course_id (PK), title	Parent of Topic
Topic	topic_id (PK), course_id (FK), name, prerequisite topic IDs	Parent of ContentItem, Question

Entity	Key attributes	Belongs to / links
ContentItem	content_id (PK), topic_id (FK), type, difficulty 1 to 5, duration	Appears in Recommendation, InteractionEvent
Question	question_id (PK), topic_id (FK), difficulty, correct answer	Answered in Attempt
Attempt	attempt_id (PK), user_id (FK), question_id (FK), correct, time taken, answered_at	Feeds MasteryScore
MasteryScore	user_id + topic_id (composite PK), score, confidence, updated_at	One row per learner and topic
InteractionEvent	event_id (PK), user_id (FK), content_id (FK), event type, duration, timestamp	Source of engagement features
Recommendation	rec_id (PK), user_id (FK), content_id (FK), arm_id (FK), rank, reason, shown_at, outcome	Outcome: accepted, dismissed or ignored
Experiment	experiment_id (PK), name, primary metric, status, start and end	Parent of Arm
Arm	arm_id (PK), experiment_id (FK), strategy, parameters, allocation share	Used by Recommendation
ExperimentAssignment	user_id + experiment_id (composite PK), arm_id (FK), assigned_at	Fixes a learner's arm

5.2 Relationships

- Course 1 to N Topic; Topic 1 to N ContentItem; Topic 1 to N Question.
- User 1 to N Attempt; Question 1 to N Attempt.
- User N to M Topic through MasteryScore.
- User 1 to N Recommendation; ContentItem 1 to N Recommendation.

- Experiment 1 to N Arm; User N to M Experiment through ExperimentAssignment.

5.3 Data rules

- A learner's personal data shall be deleted or anonymized on request.
- Events older than a configurable retention period shall be aggregated, then removed. The period is an open question for the institution.

6. Non-functional requirements

Sizing figures marked (assumed) are starting points to confirm with the institution.

ID	Category	Requirement	Verified by
NFR-01	Performance	A recommendation request shall return within 500 ms at the 95th percentile with 1,000 concurrent learners (assumed).	Load test
NFR-02	Performance	The instructor dashboard shall load within 3 seconds for a class of 500 learners.	Load test
NFR-03	Performance	A mastery update shall be visible to the next request within 1 second of the attempt.	Integration test
NFR-04	Scalability	The API and services shall scale horizontally; the system shall support 10,000 registered learners (assumed).	Load test, review
NFR-05	Security	Passwords shall be stored using a salted adaptive hash; sessions shall expire after inactivity.	Review, security test
NFR-06	Security	The system shall enforce role-based access on every endpoint and validate all input against the OWASP Top 10 risks.	Security test
NFR-07	Privacy	The system shall collect only data needed for recommendations, honor deletion requests and comply with applicable data-protection law.	Audit

ID	Category	Requirement	Verified by
NFR-08	Reliability	Monthly availability shall be at least 99.5% (assumed); if a strategy fails, the system shall fall back to the rule-based strategy.	Monitoring, fault test
NFR-09	Reliability	Data shall be backed up daily and restorable.	Restore drill
NFR-10	Usability	A new learner shall complete the diagnostic quiz without instructions; the UI shall meet WCAG 2.1 AA.	Usability test, audit
NFR-11	Explainability	Every recommendation shall show a reason a learner can read in under 5 seconds.	Demonstration
NFR-12	Maintainability	New strategies shall be added by implementing one interface, with no change to clients; core logic shall have at least 70% unit-test coverage.	Code review, coverage report
NFR-13	Configurability	Thresholds, weights and learning rates shall be changeable without redeploying.	Demonstration
NFR-14	Portability	The system shall run in containers on any Linux host.	Deployment test

7. Algorithms and models

The MVP uses simple, explainable formulas. Values below are initial defaults, tunable by administrators.

7.1 Mastery update

After each graded attempt on a question of difficulty d (scaled 0 to 1), the learner's mastery m for that topic moves toward the observed outcome y (1 correct, 0 incorrect).

```
p_t = \sigma\big(\beta\,(m_t - d)\big), \qquad m_{t+1} =
\mathrm{clip}\big(m_t + \eta\,(y_t - p_t),\,0,\,1\big)
```

Idle topics decay with time since last practice:

```
m(\Delta t) = m_{\text{last}}\; e^{-\lambda\,\Delta t}
```

7.2 Recommendation ranking

Each eligible content item i is scored; the top N are returned with a reason.

```
S(i) = w_1\,\text{gap}(i) + w_2\,\text{fit}(i) + w_3\,\text{ready}(i) + w_4\,\text{eng}(i)
```

Term	Meaning	Range
gap(i)	1 minus mastery of the item's topic: bigger for weaker topics	0 to 1
fit(i)	Closeness of item difficulty to the learner's mastery plus a small stretch	0 to 1
ready(i)	Lowest mastery among the item's prerequisite topics	0 to 1
eng(i)	Learner's historical completion rate for this content type	0 to 1

7.3 Adaptive difficulty

Target difficulty is the current mastery plus a stretch: d* = clip(m + δ, 0, 1). Step the target up one level after consecutive correct answers and down one level after consecutive incorrect ones.

Parameter	Symbol	Default
Learning rate	η	0.10
Sensitivity slope	β	4
Decay rate per day	λ	0.01
Ranking weights	w1, w2, w3, w4	0.4, 0.3, 0.2, 0.1
Stretch	δ	0.10
Step up after	correct in a row	3
Step down after	incorrect in a row	2

For the embeddings and collaborative strategies (FR-507, FR-508), the similarity score replaces or is blended into fit(i); the interface and output stay the same.

8. Use cases and acceptance criteria

UC-01 Take the diagnostic quiz (FR-601, FR-303)

- **Actor:** learner. **Precondition:** enrolled in a course, consent given.
- **Main flow:** 1) Learner starts the diagnostic. 2) System serves questions across all topics. 3) Learner answers. 4) System stores attempts and sets initial mastery per topic. 5) System shows the first recommended path.
- **Acceptance:** Given a new learner, when the diagnostic ends, then every topic has a mastery score and at least one recommendation is shown.

UC-02 Follow the recommended path (FR-501 to FR-505)

- **Actor:** learner. **Precondition:** at least one mastery score exists.
- **Main flow:** 1) Learner opens the course home. 2) System returns the top N items with reasons. 3) Learner opens an item; the system logs the event. 4) Learner may dismiss an item (FR-509).
- **Alternate:** no history exists, so the fallback in FR-510 applies.
- **Acceptance:** Given weak topic T, when the learner requests recommendations, then the list includes items for T, excludes items with weak prerequisites, and each item shows a reason.

UC-03 Take an adaptive quiz (FR-602 to FR-605)

- **Actor:** learner.
- **Main flow:** 1) System selects a question at the target difficulty. 2) Learner answers. 3) System updates mastery and the next target difficulty. 4) Repeat until the stop rule fires.
- **Acceptance:** Given three correct answers in a row, the next question is one difficulty level higher; given two incorrect in a row, one level lower; no question repeats in a session.

UC-04 Review at-risk learners (FR-702, FR-703)

- **Actor:** instructor.
- **Main flow:** 1) Instructor opens the class dashboard. 2) System shows a topic heatmap and an at-risk list. 3) Instructor opens one learner's history.
- **Acceptance:** Given a learner who meets the configured at-risk criteria, the learner appears in the list; an instructor cannot view learners outside their own courses.

UC-05 Run an A/B experiment (FR-801 to FR-806)

- **Actor:** administrator.
- **Main flow:** 1) Admin defines arms, split and primary metric. 2) System assigns learners at random, once. 3) System logs recommendations per arm. 4) Admin reads per-arm metrics with confidence intervals.
- **Acceptance:** Given a 50/50 split, each learner stays in one arm for the whole experiment, and a guardrail breach pauses it automatically.

9. Traceability matrix

Every PRD user story maps to requirements and a verification method.

PRD story	Functional requirements	Non-functional	Verification
US-01 Diagnostic quiz	FR-601, FR-303, FR-510	NFR-10	UC-01 acceptance test
US-02 Strength per topic	FR-301, FR-302, FR-306, FR-604, FR-701	NFR-03	Unit tests on mastery update; dashboard check
US-03 Recommended next lesson	FR-501 to FR-505, FR-305, FR-307	NFR-01, NFR-11	UC-02 acceptance test; load test
US-04 Adaptive quizzes	FR-602, FR-603, FR-605	NFR-03	UC-03 acceptance test
US-05 Dismiss a recommendation	FR-509	None	Functional test
US-06 Streaks and time spent	FR-307, FR-701	None	Dashboard check
US-07 Class mastery and at-risk	FR-702, FR-704, FR-103	NFR-02	UC-04 acceptance test
US-08 Learner drill-down	FR-703, FR-103	NFR-06	Access-control test
US-09 Content tagging	FR-201, FR-202, FR-203	None	Functional test
US-10 A/B experiments	FR-506, FR-801 to FR-806	NFR-12	UC-05 acceptance test
US-11 Strategy switch	FR-807	NFR-13	Demonstration

PRD story	Functional requirements	Non-functional	Verification
US-12 Export report	FR-705	None	Functional test